

AWS CloudTrail

05-monitoring/cloudtrail

Overview

CloudTrail records AWS API activity. It is the audit log of who called what, from where, and when across your AWS accounts.

This matters in ML platforms because operational and security incidents often revolve around control plane changes: a role was broadened, a training job was launched, an endpoint was deleted, or a bucket policy changed.

Core Concepts

- Management events cover control plane actions such as creating resources or changing permissions. Data events cover high volume object or item level activity for selected services such as S3.
- Trails can be account scoped or organization wide, and their logs are usually delivered to S3 with optional integration into analytics or alerting systems.
- CloudTrail Lake and event history features provide additional ways to search and analyze activity, but the core value remains reliable auditability.

How It Works

- A user, role, or service calls an AWS API. CloudTrail records the event metadata and writes it into the configured delivery and retention path.
- Security teams or operators query those logs to answer investigative questions such as who changed this policy, which role deleted the endpoint, or when access started failing.
- CloudTrail records can also drive detections when sensitive actions occur outside expected automation paths.

AI and ML Relevance

- CloudTrail helps prove whether a model deployment was automated, whether a dataset bucket was accessed unexpectedly, or whether a training cluster was launched by a human rather than by the approved pipeline.
- It is also useful for governance because regulated ML workloads need change history around access, release, and data handling operations.

Operational Notes

- Enable organization wide coverage where possible, and protect the destination logs from tampering with strong access controls and retention design.
- Data events can be high volume and higher cost, so enable them where their investigative value is worth the overhead.
- CloudTrail is most useful before an incident, not after. Missing audit coverage cannot be reconstructed retroactively.

What To Remember

- CloudWatch tells you a system is unhealthy; CloudTrail often tells you what changed.
- It is an audit and governance service, not a performance monitoring tool.
- In mature ML environments, every production change path should be traceable through CloudTrail records.